

Mid Semester Examination CS201

Time Duration: 2 hours

Instructions:

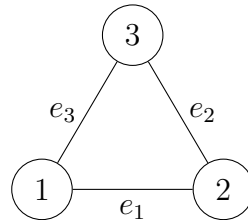
1. For Questions 1–8, only the final answers are required. Do your rough work separately (otherwise marks may not be awarded).
2. For the remaining questions, write brief and to-the-point arguments, unless explicitly stated otherwise (see Questions 14(a) and 17(a)).

Questions:

1. (0.5+1 marks) Let $S(n, k)$ be the stirling number of second kind. Write the values of $S(5, 5)$, $S(8, 2)$.

1, $2^7 - 1 = 127$

2. (1 marks) For the following graph, write down (i) the adjacency matrix, and (ii) the incidence matrix.



3. (1 marks, marks will be provided only if you choose all correct answers)
Which of the following relations on the set $\{1, 2, 3\}$ is a partial order?

- (a) $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$
- (b) $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$
- (c) $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$
- (d) $R = \{(1, 1), (2, 2), (3, 3)\}$

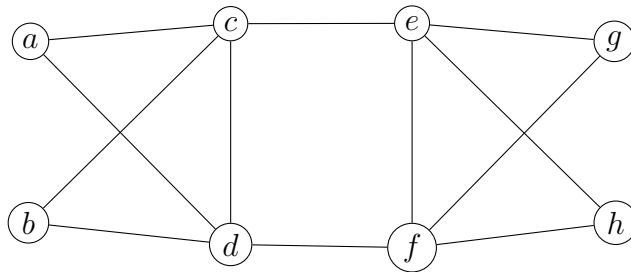
A,C,D

4. (1 marks, marks will be provided only if you choose all correct answers)
In the above question, which of the relations is an equivalence relation?

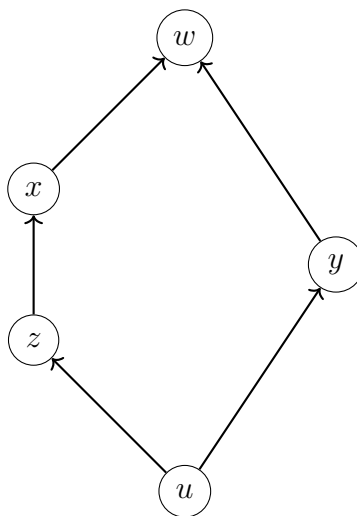
B,D

5. (1 marks) Find an Eulerian tour in this graph:

$c \rightarrow d \rightarrow a \rightarrow c \rightarrow b \rightarrow d \rightarrow f \rightarrow h \rightarrow e \rightarrow f \rightarrow g \rightarrow e \rightarrow c$



6. (1 marks) Consider a pentagon lattice whose Hasse diagram is shown below:



Show this lattice is not distributive by giving values of $a, b, c \in \{u, z, y, x, w\}$ such that

$$a \wedge (b \vee c) \neq (a \wedge b) \vee (a \wedge c).$$

$a = x$, and one of b, c will be y other will be z

7. (1 marks) Let $a_n - 3a_{n-1} + 2a_{n-2} = 0$ for $n \geq 2$ with $a_0 = 1, a_1 = 2$. Find the OGF $A(x)$ for the sequence $a_0, a_1, a_2, \dots$.

$$\frac{1}{1-2x} = \frac{1-x}{1-3x+2x^2}$$

8. (1 marks) Let $a_{n+1} = (n+1)a_n + 1$ for $n \geq 0$, with $a_0 = 1$. Derive the EGF $A(x)$ for the sequence $a_0, a_1, a_2, \dots$.

$$\frac{e^x}{1-x}$$

9. (1 marks) Can a bipartite graph with odd number of vertices have a Hamiltonian cycle? Justify.

No. By definition, a Hamiltonian cycle will include all the vertices. If the number of vertices is odd then there will exist a cycle in the graph of odd length. By definition a graph is bipartite if and only if it has no odd cycles.

10. (1 marks) Let G is a simple connected graph with at least two vertices, prove that G must contain two or more vertices of the same degree.

By pigeonhole principle. Since the graph is connected, all vertices has degree at least 1. The possible values of degree are $1, 2, \dots, n-1$ (total $n-1$ possible values). Apply pigeonhole principle, with set of Pigeonholes as $\{1, 2, \dots, n-1\}$ and vertex set as set of pegions, assign vertex v to j if degree of v is j .

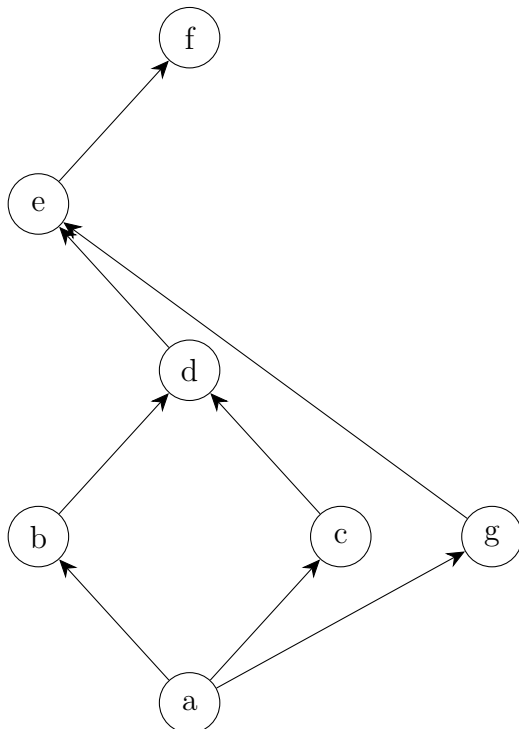
11. (1.5 marks) In a poset with $ab+1$ elements, show that there will either a antichain of size $a+1$ or a chain of size $b+1$.

If there exists an antichain of size $a+1$ then we are done. So assume size of each antichain is at most a . So at least $b+1$ antichains will be needed to cover the post. By Dilowrth theorem, there exists a chain of size at least $b+1$.

12. (1.5 marks) Prove that $\log_2 3$ is irrational.
let $\log_2 3 = p/q$ then $2^p = 3^q$. one is even other is odd
13. (3 marks) Prove by induction that any natural number $n \geq 12$ can be expressed as $3x + 7y$ for non negative integers x and y .
Base case: $n=12,13,14$. Then show that $P(n) \implies P(n+3)$ where $P(n)$ is the statement in question to be proven.
14. (1+2+2 marks) Consider a poset whose Hasse diagram is shown below:
- (a) (no explanation needed) Determine all the element(s) that are the upper bounds on the set $\{b, c, g\}$ and all the element(s) that are lower bound on the set $\{b, c, g\}$.
- upper bounds: e, f , Lower bounds: a**
- (b) Verify Dilworth's Theorem (as well as its dual version) for this poset.
Maximum size of antichain = 3. Poset can be also be partitioned into 3 chains: $C_1 = \{a, b, d, e, f\}, C_2 = \{c\}, C_3 = \{g\}$. Similarly, maximum size of chain is 5. Poset can be partitioned into 5 antichaig: $\{a\}, \{b, c, g\}, \{d\}, \{e\}, \{f\}$

(c) Is this poset a lattice? Justify

Yes. In any poset if $a \leq b$ then lub of $\{a, b\}$ is b and glb on $\{a, b\}$ is a . So it suffices to check only for incomparable pair of elements which are $\{b, c\}, \{b, g\}, \{c, g\}, \{d, g\}$.



15. (3 marks) Consider any lattice $(\mathcal{L}, \leq)$. Show that if $a \leq b$ then $a \vee (x \wedge b) \leq (a \vee x) \wedge b$.

$a \leq (a \vee x)$ (as $a \vee x$ is an upper bound on a) and $a \leq b$ (given)
 thus a is a lower bound on $\{(a \vee x), b\}$ and since any lower bound is less than or equal to the greatest lower bound we have $a \leq (a \vee x) \wedge b$ (because $\wedge$ is the greatest lower bound)

So we got $a \leq (a \vee x) \wedge b$.

Similarly, we will show that $x \wedge b \leq (a \vee x) \wedge b$. Clearly $x \wedge b \leq b$ and $x \wedge b \leq x \leq (a \vee x)$. Again $x \wedge b$ is a lower bound on $\{b, a \vee x\}$.

Since any lower bound is less than or equal to the greatest lower bound, we have $x \wedge b \leq (a \vee x) \wedge b$.

Now we have shown that both a and $x \wedge b$ is $\leq (a \vee x) \wedge b$. In other words, $(a \vee x) \wedge b$ is an upper bound on $\{a, x \wedge b\}$. Since least upper bound is less than or equal to any upper bound so we have $a \vee (x \wedge b) \leq (a \vee x) \wedge b$.

16. (3 marks) Consider a graph G that has exactly two connected components. Show that the number of edges in G is at most $(n-1)(n-2)/2$.

Let one component has a vertices and other has b vertices. The maximum possible edges with this condition is $\binom{a}{2} + \binom{b}{2}$. Thus maximum possible edges over all graph is

$$\max_{a,b:a+b=n} \binom{a}{2} + \binom{b}{2}$$

This maximum will be attained when $a = 1$ and $b = n - 1$ (can be shown in many ways - for example show that above function value at a, b is at most value at $a - 1, b + 1$). In this case, maximum number of edges is $(n-1)(n-2)/2$.

17. Let $c(n, k)$ be the number of permutations on n items with exactly k cycles.

(a) (1 marks, no expalnation needed) Write the value of $c(n, n - 1)$.

$\binom{n}{2} = \frac{n(n-1)}{2}$. Exactly one cycle will be of size 2 and rest will be of size 1. There are $\binom{n}{2}$ ways to choose two elements that form a cycle.

(b) (2 marks) Recurrence for $c(n, k)$: write $c(n, k)$ in terms of $c(n - 1, k - 1)$ and $c(n - 1, k)$ and prove your claim.

$c(n, k) = c(n - 1, k - 1) + (n - 1)c(n - 1, k)$. Either 1 will get mapped to itself (total number of such permutations is $c(n - 1, k - 1)$) or 1 will not get mapped to itself. In the later case, given a permutation of $\{2, 3, \dots, n\}$ with k cycles, element 1 can be added in $n - 1$ ways.

(c) (3 marks) Show that

$$\sum_{k=1}^n c(n, k)x^k = x(x+1) \cdots n \text{ terms} = x(x+1)(x+2) \cdots (x+n-1)$$

can be shown using the above recurrence and initial conditions $c(n, 1) = (n-1)!$ and by induction:

$$\begin{aligned} \sum_{k=1}^n c(n, k)x^k &= c(n, 1)x + \sum_{k=2}^n (c(n-1, k-1) + (n-1)c(n-1, k))x^k \\ &= c(n, 1)x + x \sum_{k=2}^n c(n-1, k-1)x^{k-1} + (n-1) \sum_{k=2}^n c(n-1, k)x^k \\ &= c(n, 1)x + x \sum_{k=1}^{n-1} c(n-1, k)x^k + (n-1) \left(\sum_{k=1}^n c(n-1, k)x^k - c(n-1, 1)x \right) \\ &= \left(\sum_{k=1}^{n-1} c(n-1, k)x^k \right)(x+n-1) + c(n, 1)x - (n-1)c(n-1, 1)x \\ &= x(x+1) \cdots (x+n-2)(x+n-1) \text{ (by induction and by } c(n, 1) = (n-1)!) \end{aligned}$$

18. (3 marks) Let n be even. Let c_n be the number of permutations in which all cycles are of even length. Write c_n in terms of c_{n-2} for $n \geq 4$ (and prove your claim).

$c_n = (n-1)c_{n-2} + (n-1)(n-2)c_{n-2} = (n-1)^2 c_{n-2}$. **Either the cycle containing 1 will be of size 2 or at least 4. In the first case, one cycle in the permutation will be $1 \rightarrow i \rightarrow 1$. The values of i can be chosen in $n-1$ ways. Total number of permutations with a fixed value of i is c_{n-2} . Thus total number of all such permutations is $(n-1)c_{n-2}$. In the later case, 1 can get mapped to i which can get mapped to j (total possibilities of (i, j) is $(n-1)(n-2)$). Now let us count total permutations with fixed value of i and j . Wlog $i = 2, j = 3$. Now consider $(1 \rightarrow 2 \rightarrow 3)$ as one unit, i.e., think of it as another element say $n+1$. Now given any permutations on $\{4, 5, \dots, n, n+1\}$ with even cycles, we can replace element $n+1$ by $1 \rightarrow 2 \rightarrow 3$, i.e., we replace $a \rightarrow n+1 \rightarrow b$ by $a \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow b$ (length of cycle will**

increase by 2 so it still be even) to get a permutation with our requirement. This total number of such permutations is c_{n-2}

19. (3 marks) Fifteen chairs are evenly placed around a circular table. On the table are the name cards of fifteen guests, in front of every chair. After the guests sit down, it turns out that none of them is sitting in front of his own card. Prove that there exists a rotation of the table that places at least two guests in front of their correct cards.

Application of pigeon hole principle. Pigeons are 15 guests and pigeonholes are 14 configurations : all configurations obtained by rotation of table minus the initial configuration. If in configuration C , guest i sits correctly then assign guest i to C . Note that for every guest, there will exist a configuration in which he/she sits correctly. In the initial configuration, no guest sits on the correct position so by pigeonhole principle, at least 2 guests will get mapped to the same configuration. In other words, there exists a configuration such that at least 2 guests will sit correctly.